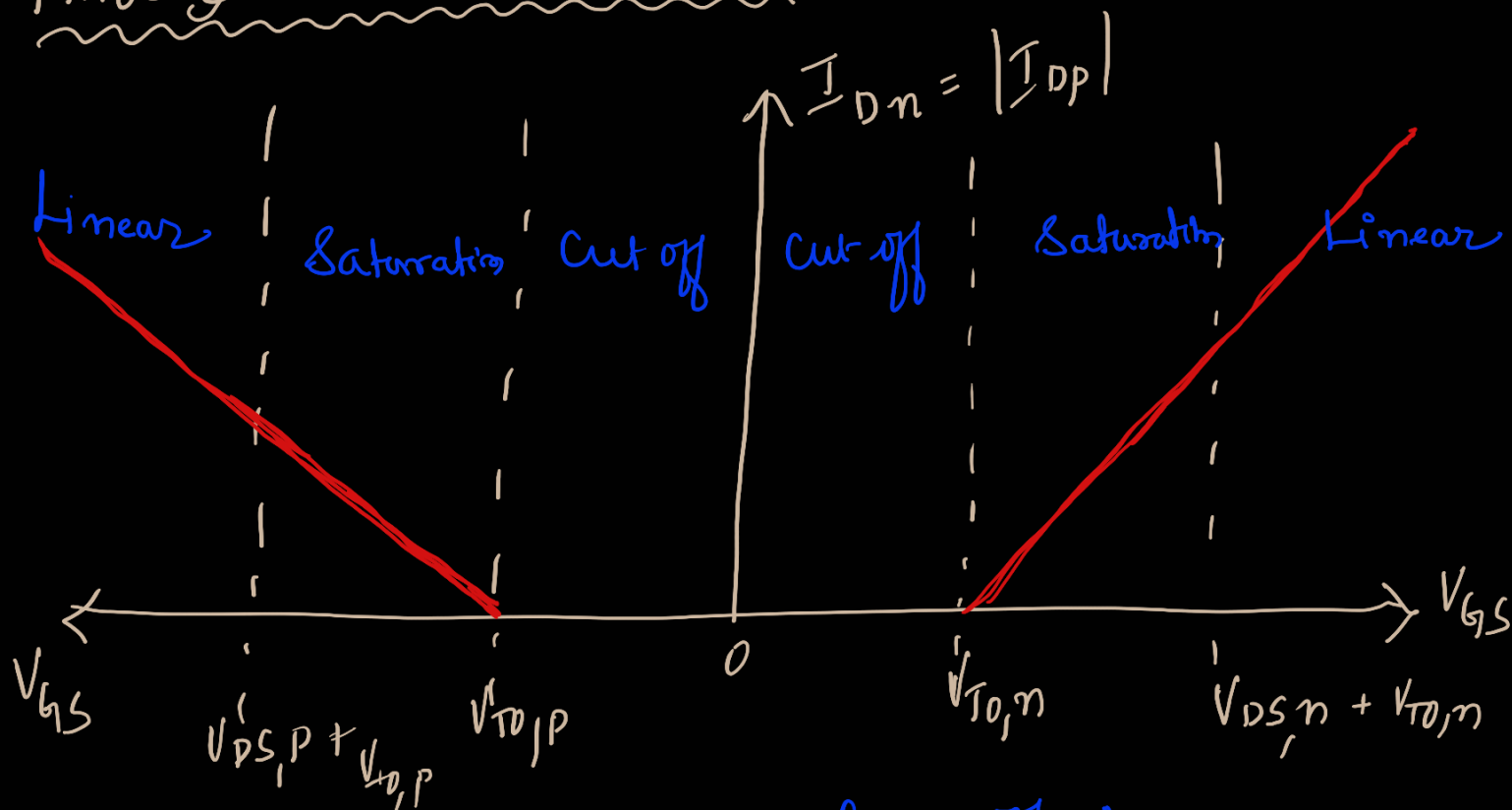


CMOS Inverter

PMOS & NMOS Characteristics



Cut off:

$$V_{GS,p} > V_{to,p}$$

Cut off:

$$V_{GS,n} < V_{to,n}$$

Saturation:

$$V_{GS,p} < V_{to,p}$$

$$V_{DS,p} \leq V_{GS,p} - V_{to,p}$$

Saturation:

$$V_{GS,n} > V_{to,n}$$

$$V_{DS,n} \geq V_{GS,n} - V_{to,n}$$

Linear:

$$V_{GS,p} < V_{to,p}$$

$$V_{GS,p} > V_{GS,p} - V_{to,p}$$

Linear:

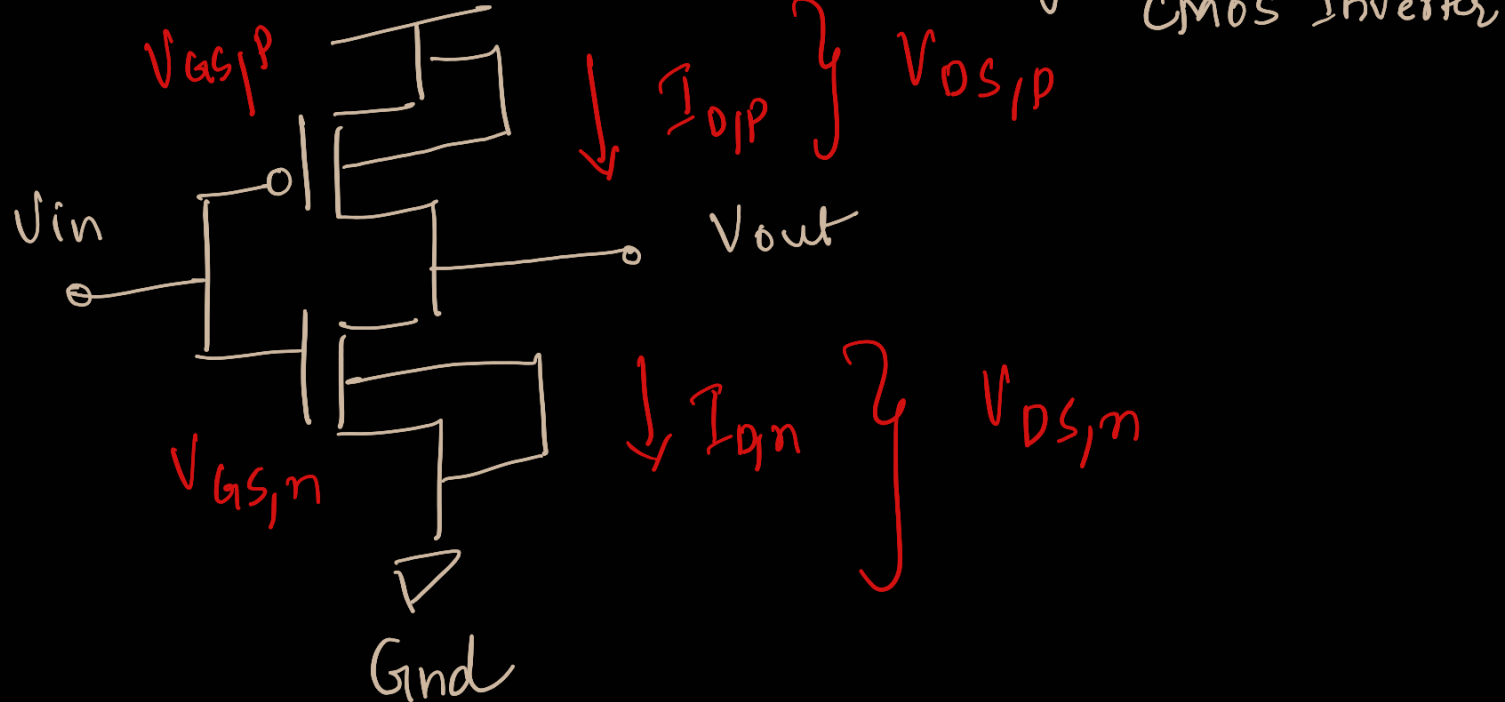
$$V_{GS,n} < V_{to,n}$$

$$V_{DS,n} < V_{GS,n} - V_{to,n}$$

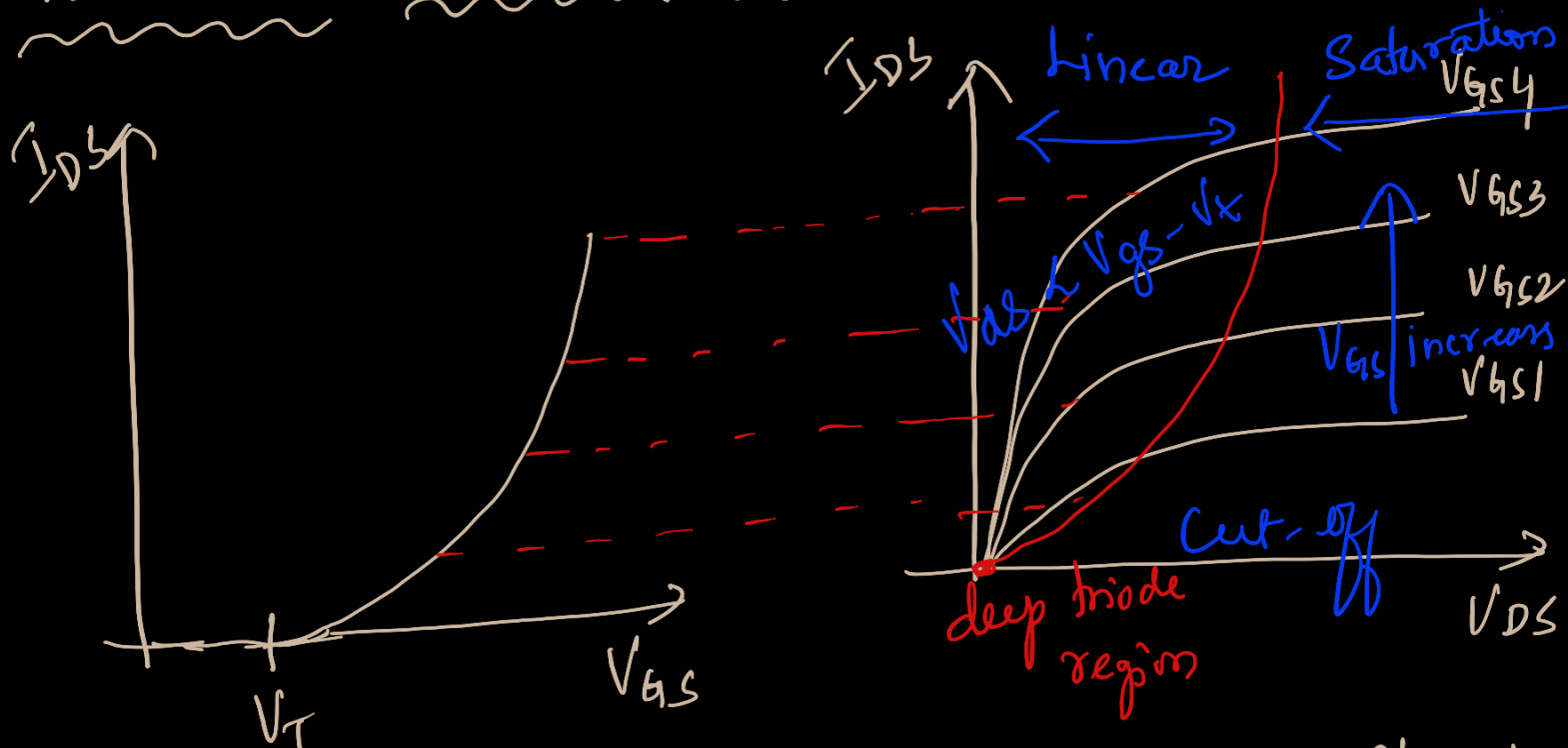
CMOS Inverter:

V_{DD}

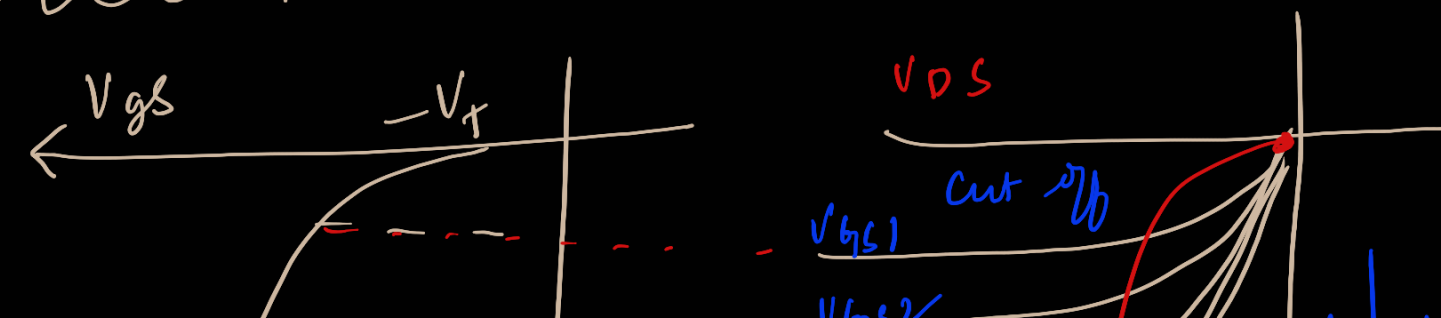
Fig: Schematic of a CMOS Inverter

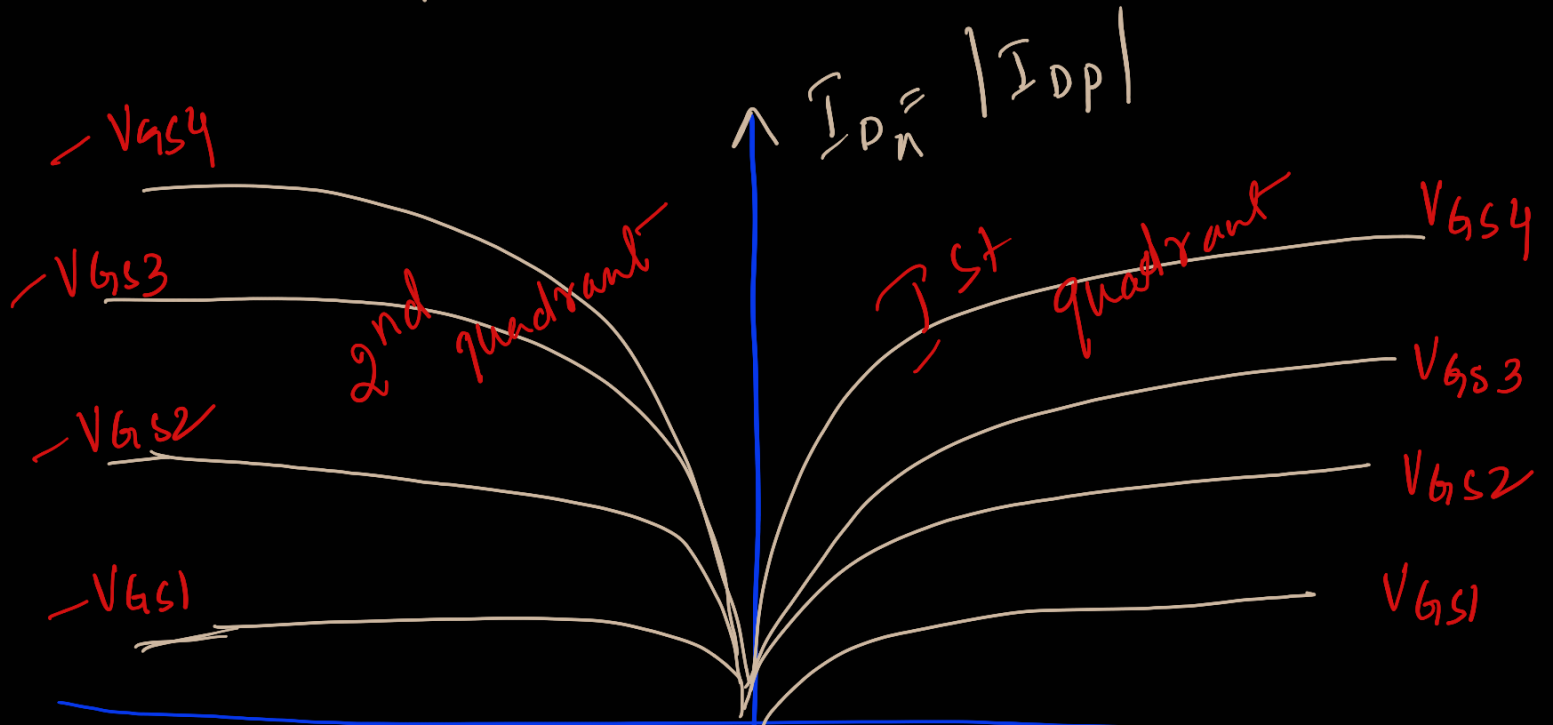
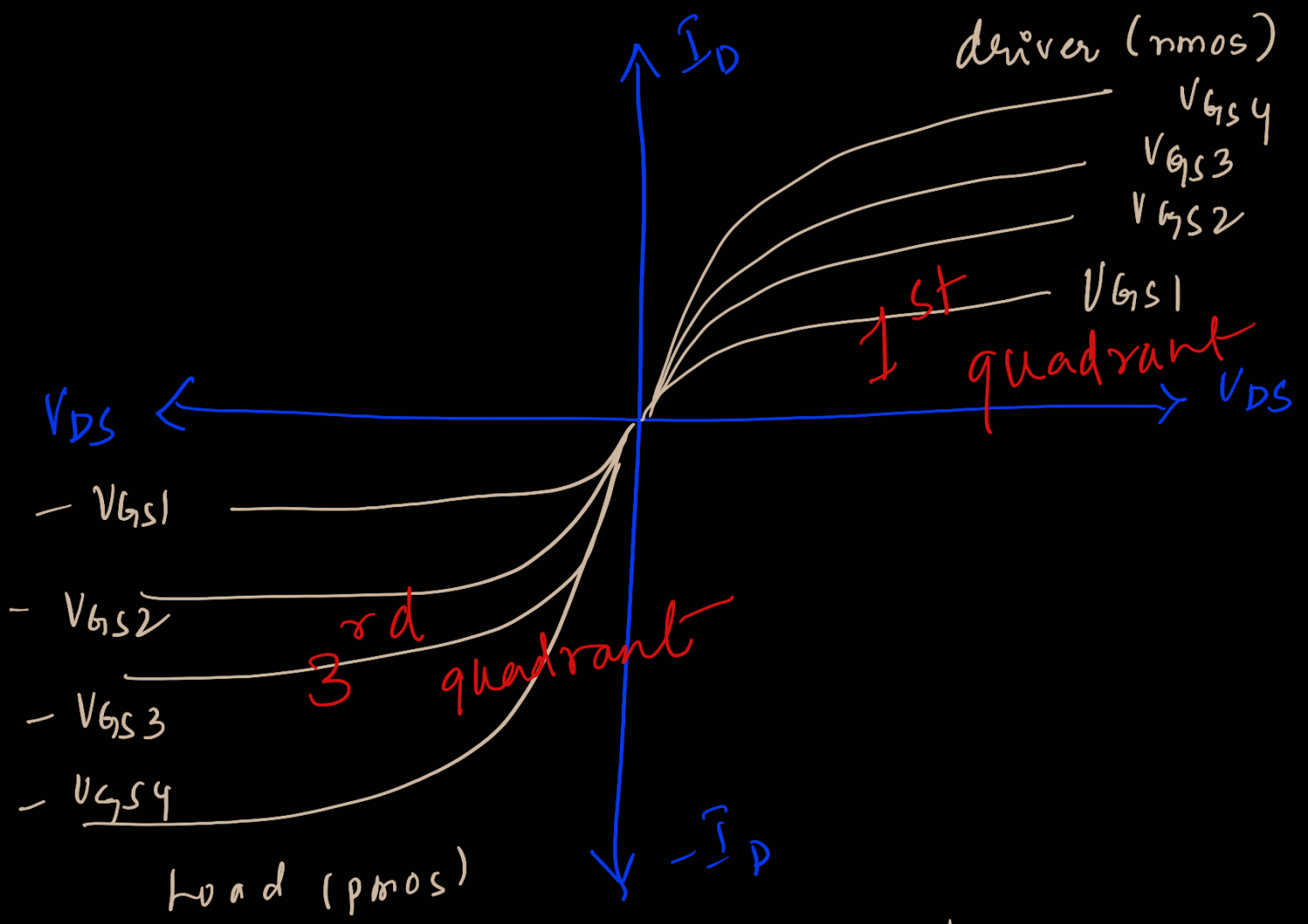


n-channel Enhancement Type MOSFET



p-channel Enhancement Type MOSFET

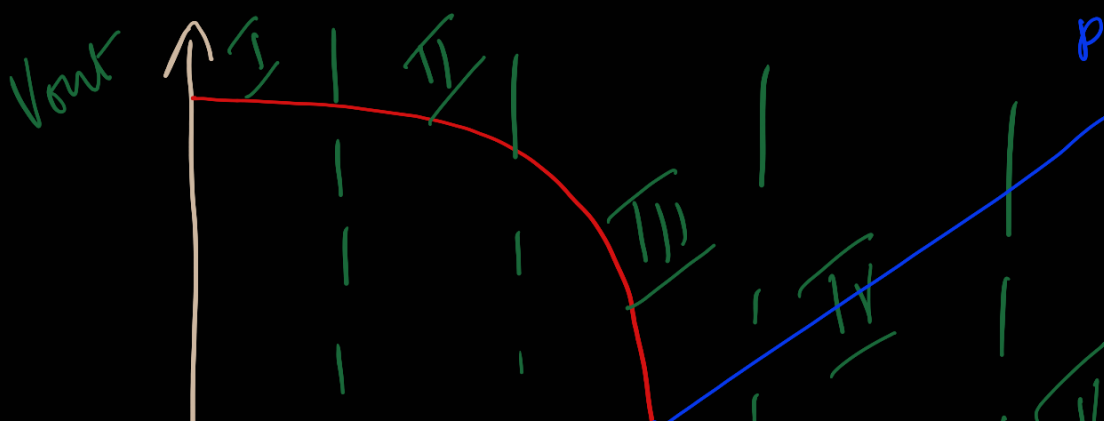
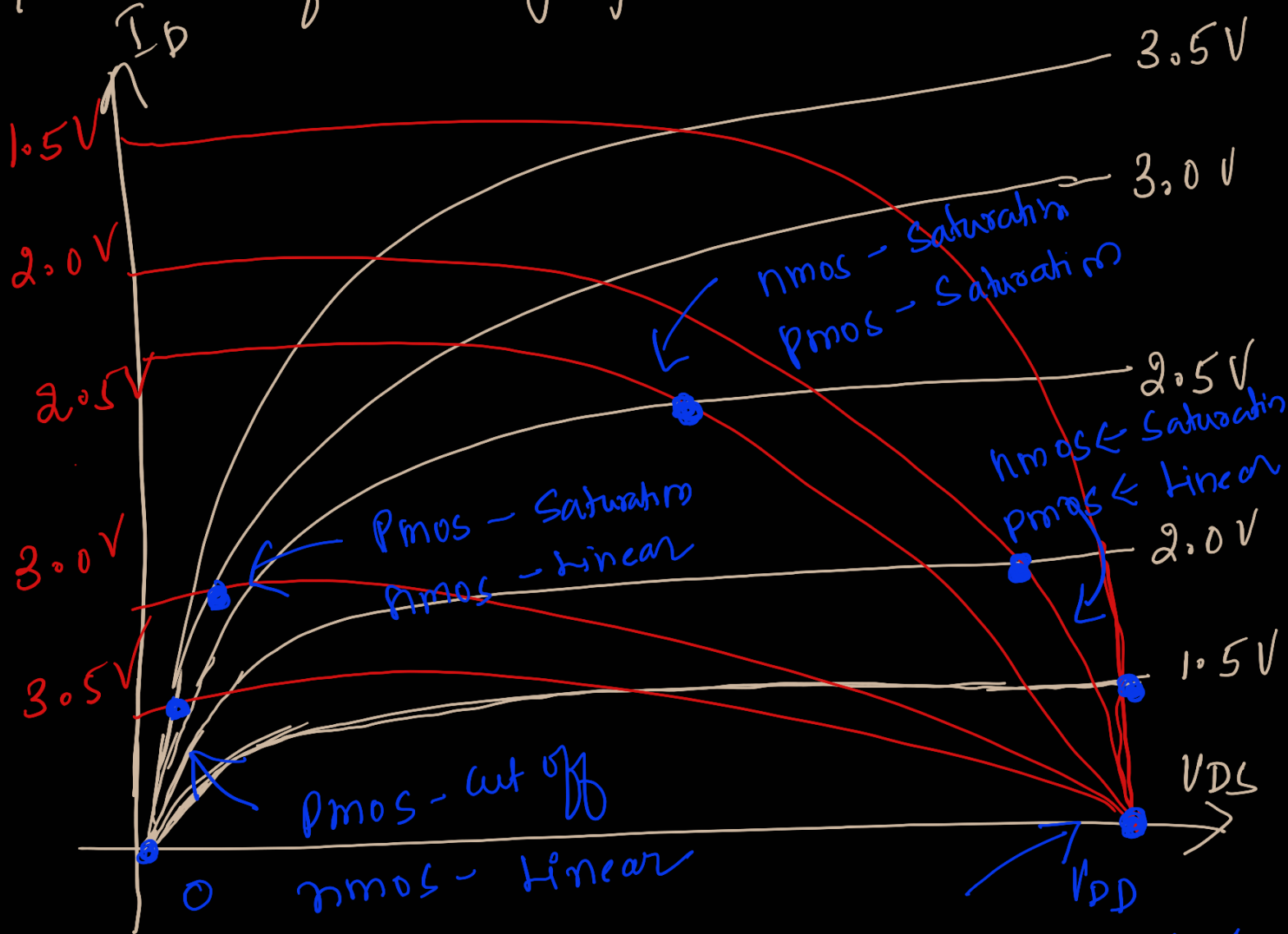


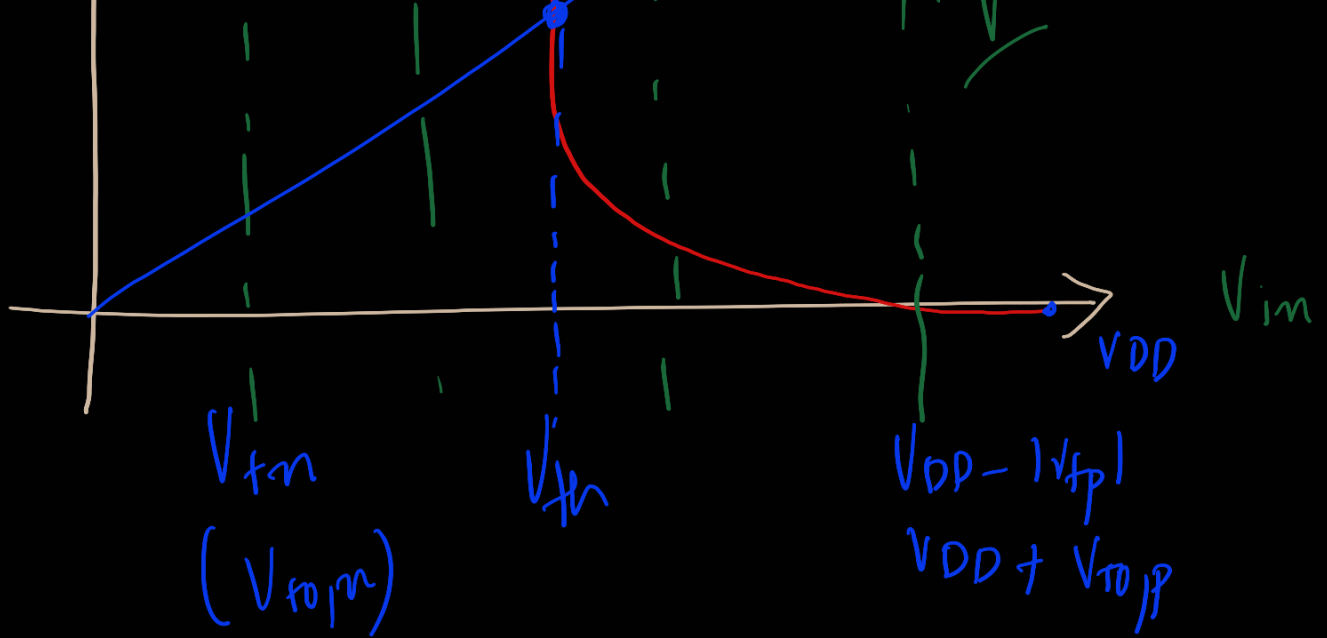


V_{DS}

V_{DS}

Impinge the output characteristics of pmos (2nd quadrant of the graph) on the output characteristics of nmos (1st quadrant of the graph).





Regions

- (A) I : $V_{in} < V_{to,n}$: $V_{out} = V_{DD}$
 (B) II : PMOS $\rightarrow$ Linear, NMOS $\rightarrow$ Saturation
 (C) III : PMOS & NMOS $\rightarrow$ Saturation
 (D) IV : NMOS $\rightarrow$ Linear, PMOS $\rightarrow$ Saturation

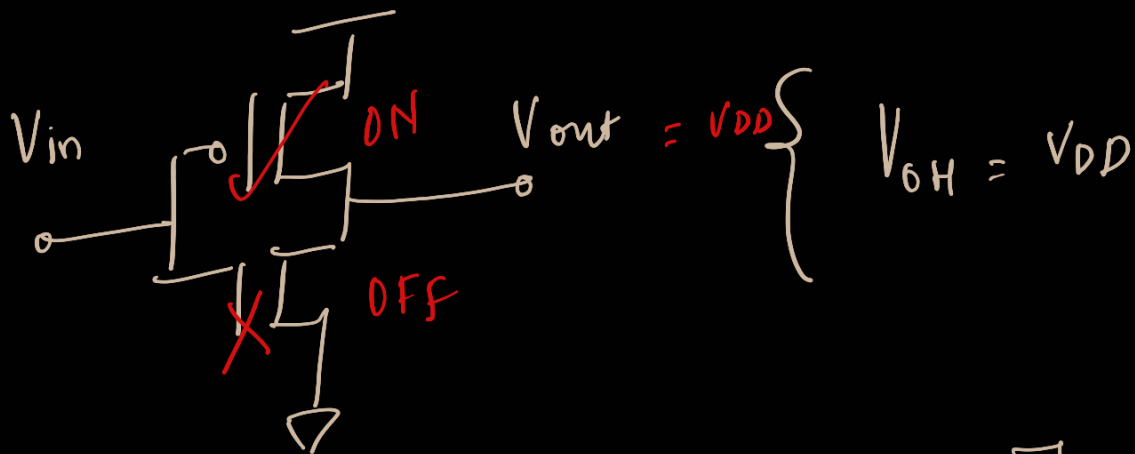
Region of Operations :

Region	V_{in}	V_{out}	nmos	pmos
A	$< V_{to,n}$	V_{OH}	Cut off	Linear
B	V_{IL}	$\approx V_{OH}$	Saturation	Linear
C	V_{th}	V_{th}	Saturation	Saturation
D	V_{IH}	$\approx V_{OL}$	Linear	Saturation

V_{in}	$> (V_{DD} + V_{th,p})$	V_{OL}	Linear	Cut off
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REGION A :

When $V_{in} < V_{th,n}$



REGION B : [calculation of V_{IL}]

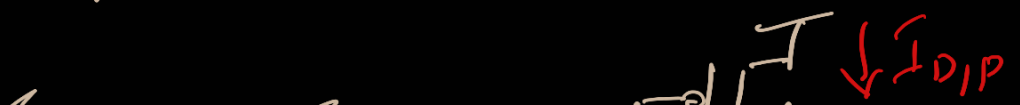
→ As V_{in} is increased beyond $V_{th,n}$, NMOS conducts in saturation and the output voltage starts to decrease

→ Critical Voltage V_{IL} is defined at

$\frac{dv_{out}}{dv_{in}} = -1$ is located within region B.

* nmos → Saturation

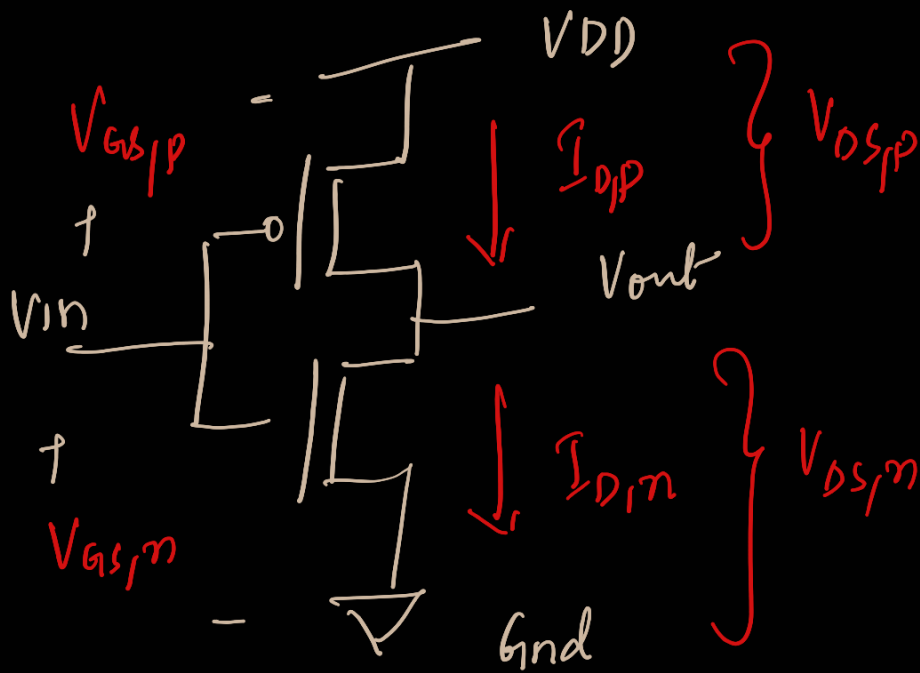
* pmos → Linear



$$I_{D,n} = I_{D,p} \quad \begin{array}{c} \downarrow \\ \downarrow \\ \downarrow \end{array} \quad \downarrow I_{D,n}$$

$$I_{D,n} = \frac{k_n}{2} (v_{gs,n} - v_{to,n})^2$$

$$I_{D,P} = \frac{K_n}{2} \left[2(V_{GS,P} - V_{TO,P}) V_{DS,P} - V_{DS,P}^2 \right]$$



$$V_{G_{S,n}} = V_{Sn}$$

$$V_{GS,p} = V_{in} - V_{DD}$$

$$V_{GS,n} = V_{out} - V_{DD}$$

$$\frac{k_n}{2} (v_{in} - v_{tpn})^2 = \frac{K_p}{2} \left[2 (v_{in} - v_{DD} - V_{tpp}) (v_{out} - v_{DD}) - (v_{out} - v_{DD})^2 \right] \rightarrow \textcircled{1}$$

$uv = u'v + uv'$ rule

As V_{IL} is defined at $\frac{dv_{out}}{dv_{in}} = -1$. In order to

Satisfy this condition, we differentiate equ no (1)
w.r.t V_{in} on both sides

$$2 K_n (v_{in} - v_{to,n}) = k_p \left[2 (v_{in} - v_{DD} - v_{to,p}) \left(\frac{dv_{out}}{dv_{in}} \right) + \right. \\ \left. + (v_{in} - v_{DD}) - 2 (v_{out} - v_{DD}) \frac{dv_{out}}{dv_{in}} \right]$$

Substitute $\frac{dv_{out}}{dv_{in}} = -1$ and $v_{in} = v_{IL}$ in equ no (2)

$$K_n (v_{IL} - v_{TO,n}) = K_p [-v_{IL} + v_{DD} + v_{TO,p} + v_{out} - v_{DD} + v_{out} - v_{DD}]$$

$$K_n (v_{IL} - v_{TO,n}) = K_p [2v_{out} - v_{IL} + v_{TO,p} - v_{DD}]$$

$$K_n v_{IL} + K_p v_{IL} = 2K_p v_{out} + K_p v_{TO,p} - K_p v_{DD} + K_n v_{TO,n}$$

$$v_{IL} (K_n + K_p) = K_p (2v_{out} + v_{TO,p} - v_{DD}) + K_n v_{TO,n}$$

$$v_{IL} = \frac{K_p (2v_{out} + v_{TO,p} - v_{DD}) + K_n v_{TO,n}}{(K_n + K_p)}$$

Taking K_p Common,

$$v_{IL} = \frac{2v_{out} + v_{TO,p} - v_{DD} + \left(\frac{K_n}{K_p}\right) v_{TO,n}}{\left(1 + \frac{K_n}{K_p}\right)}$$

Where $K_R = \frac{K_n}{K_p}$

$$v_{IL} = \frac{2v_{out} + v_{TO,p} - v_{DD} + K_R v_{TO,n}}{1 + K_R}$$

Where

$K_R =$ Transconductance parameter ratio

REGION D [Calculation of V_{IH}]

When V_{out} falls below $(V_{in} - V_{T0,n})$, the nmos transistor starts operating in linear region while pmos is still in saturation

V_{IH} is defined at $\frac{dV_{out}}{dV_{in}} = -1$ located in region D

When $V_{in} = V_{IH}$

nmos $\rightarrow$ Linear Region
pmos $\rightarrow$ Saturation Region

$$I_{D,p} = I_{D,n}$$

$$\frac{K_n}{2} \left[2(V_{gs,n} - V_{T0,n}) V_{DS,n} - V_{DS,n}^2 \right] = \frac{K_p}{2} \left[V_{gs,p} - V_{T0,p} \right]^2$$

$$V_{in} = V_{gs,n}, \quad V_{DS,n} = V_{out}, \quad V_{gs,p} = V_{in} - V_{DD}$$

$$\frac{K_n}{2} \left[2(V_{in} - V_{T0,n}) V_{out} - V_{out}^2 \right] = \frac{K_p}{2} \left[V_{in} - V_{DD} - V_{T0,p} \right]^2$$

As V_{IH} is defined at $\frac{dV_{out}}{dV_{in}} = -1$, differentiate the above equation w.r.t to V_{in}

$$K_n \left[2(V_{in} - V_{T0,n}) \left(\frac{dV_{out}}{dV_{in}} \right) + 2V_{out} (1) - 2V_{out} \left(\frac{dV_{out}}{dV_{in}} \right) \right] = K_p 2 \left[V_{in} - V_{DD} - V_{T0,p} \right]$$

Substitute $\left(\frac{dv_{out}}{dv_{in}}\right) = -1$ and $v_{in} = v_{IH}$

$$K_n \left[(V_{IH} - V_{TO,n}) (-1) + V_{out} - V_{out} (-1) \right] = K_p [V_{IH} - V_{DD} - V_{TO,p}]$$

$$K_n [-V_{IH} + V_{TO,n} + V_{out} + V_{out}] = K_p [V_{IH} - V_{DD} - V_{TO,p}]$$

$$K_p V_{IH} + K_n V_{IH} = K_n [2V_{out} + V_{TO,n}] + K_p [V_{DD} + V_{TO,p}]$$

$$V_{IH} = \frac{K_n [2V_{out} + V_{TO,n}] + K_p [V_{DD} + V_{TO,p}]}{[K_n + K_p]}$$

$$= \frac{\left(\frac{K_n}{K_p}\right) [2V_{out} + V_{TO,n}] + [V_{DD} + V_{TO,p}]}{\left(1 + \frac{K_n}{K_p}\right)}$$

$$V_{IH} = \frac{V_{DD} + V_{TO,p} + K_R (2V_{out} + V_{TO,n})}{(1 + K_R)}$$

REGION C : [Calculation of V_{th}]

→ The inverter threshold is defined as

$$V_{th} = V_{in} = V_{out}$$

→ For $V_{in} = V_{out}$, both transistors are expected to be in saturation region.

$$I_{D,n} = I_{D,p}$$

$$\frac{K_n}{2} [V_{gs,n} - V_{th,n}]^2 = \frac{K_p}{2} [V_{gs,p} - V_{th,p}]^2$$

$$V_{gs} = V_{in}, \quad V_{gs,p} = V_{in} - V_{DD}$$

$$(V_{in} - V_{th,n})^2 = \frac{K_p}{K_n} (V_{in} - V_{DD} - V_{th,p})^2$$

Taking square root on both sides

$$V_{in} - V_{th,n} = \pm \sqrt{\frac{K_p}{K_n}} (V_{in} - V_{DD} - V_{th,p})$$

The correct solution can be obtained by taking
-ve sign

$$V_{in} - V_{th,n} = - \sqrt{\frac{K_p}{K_n}} V_{in} + \sqrt{\frac{K_p}{K_n}} V_{DD} + \sqrt{\frac{K_p}{K_n}} V_{th,p}$$

$$V_{in} + \sqrt{\frac{K_p}{K_n}} V_{in} = V_{th,n} + \sqrt{\frac{K_p}{K_n}} (V_{DD} + V_{th,p})$$

$$V_{in} + \sqrt{\frac{K_p}{K_n}} V_{in} = V_{th,n} + \sqrt{\frac{K_p}{K_n}} (V_{DD} + V_{th,p})$$

$$V_{in} = \frac{V_{T0,n} + \sqrt{\frac{K_p}{K_n}} (V_{DD} + V_{T0,p})}{\left(1 + \sqrt{\frac{K_p}{K_n}}\right)}$$

$$V_{in} = \frac{V_{T0,n} + \sqrt{\frac{1}{K_R}} (V_{DD} + V_{T0,p})}{\left(1 + \sqrt{\frac{1}{K_R}}\right)}$$

V_{th} is the threshold voltage of CMOS Inverter. Where V_{in} crosses V_{out} curve. The value of V_{th} defines whether the inverter designed is symmetric or non-symmetric inverter.

$$\frac{K_n}{K_p} = \text{Transconductance Ratio}$$

As, transconductance ratio increases, V_{th} of the inverter decreases for a fixed value of V_{DD} , $V_{T0,n}$ & $V_{T0,p}$.

$$\text{If } V_{T0,n} \approx |V_{T0,p}| \text{ and } K_n = K_p$$

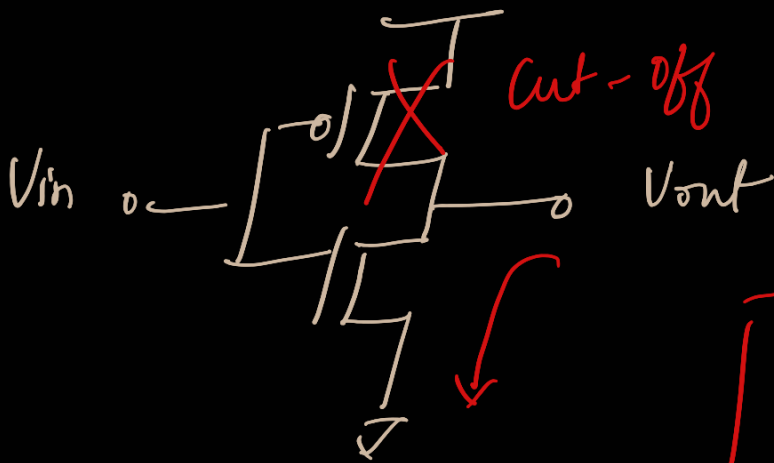
then

$$V_{in} = V_{th} = \frac{V_{DD} + V_{T0,n} + V_{T0,p}}{2}$$

$$V_{in} = V_{th} = \frac{V_{DD}}{2}$$

REGION E: (calculation of V_{OL})

→ When $V_{in} > V_{DD} + V_{TO,P}$, the pmos transistor is in cut-off region



$$V_{out} = V_{OL} = 0$$

Summary:

① $V_{OH} = V_{DD}$

② $V_{OL} = 0$

③
$$V_{IL} = \frac{2V_{out} + V_{TO,P} - V_{DD} + K_R V_{TO,n}}{(1 + K_R)}$$

④
$$V_{IH} = V_{DD} + V_{TO,P} + K_R (2V_{out} + V_{TO,n})$$

$$(1+k_R)$$

$$\textcircled{5} \quad V_{th} = \frac{V_{TO,n} + \sqrt{\frac{K_p}{K_n}} (V_{DD} + V_{TO,p})}{\left(1 + \sqrt{\frac{K_p}{K_n}}\right)}$$

